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NeuroPipe-SABV: A Data Processing Workflow Supporting Sex as a Biological Variable in Preclinical Behavioral Neuroscience

Kobe Q. Anderson [1] **and Bryan Devan**[1]

1 Laboratory for Comparative Neuropsychology, Department of Psychology, Towson University

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Software

- Review
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Summary

Behavioral neuroscience researchers often work with tables exported from animal-tracking systems. Before comparing outcomes, they must identify subjects, interpret column names and units, inspect missing values, and connect measurements to study metadata. NeuroPipe-SABV is an open-source Python application that brings these steps into a browser-based workflow, with particular attention to sex as a biological variable (SABV). It supports CSV and Excel ingestion, header standardization, user-configured sex assignment, descriptive summaries, male–female comparisons, visualization, and result export.

The repository contains a Panel interface for upload, processing, analysis, and download, alongside an earlier Streamlit interface. The Panel application can be run locally or served using the supplied Docker configuration. The software is distributed under the MIT license at https://github.com/kobeqanderson-png/NeuroPipe-SABV. Public hosting is being migrated; a continuously available hosted service is not a prerequisite for local use. NeuroPipe-SABV supports data preparation and exploratory analysis, rather than certifying NIH policy compliance or selecting a scientifically appropriate analysis on the researcher's behalf.

Statement of need

The NIH expects sex to be considered in the design, analysis, and reporting of vertebrate animal and human studies, with strong justification required for single-sex investigations (National Institutes of Health, 2015). The policy followed a broader effort to address reliance on male animals and cells (Clayton & Collins, 2014; Sandberg et al., 2015). In neuroscience, including sex can reveal biological mechanisms and behavioral strategies that would otherwise remain unexamined (Shansky, 2018; Shansky & Woolley, 2016). This requires attention to the meaning of behavioral measurements as well as the inclusion of both sexes (Shansky, 2024; Shansky & Murphy, 2021).

NeuroPipe-SABV targets researchers who receive tabular behavioral exports and need a guided route from those files to inspectable summaries. Export schemas can differ in column vocabulary, text encoding, and subject identifiers. Study-specific metadata may be maintained separately from tracking output. Repeatedly reconciling these conventions across files creates opportunities for inconsistent preprocessing and sex assignment.

The project's contribution is an application-level workflow connecting import, header mapping, explicit ID-based grouping rules, exploratory comparisons, and export. A numeric identifier does not independently reveal an animal's sex: the researcher supplies the convention and must verify assignments against study records. Similarly, producing sex-disaggregated plots does not establish that the experimental design, sample size, or inferential model satisfies the research question or NIH expectations.

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State of the field

General-purpose tools already provide the statistical operations used here. SciPy supplies statistical tests (Virtanen et al., 2020), statsmodels supports regression and hypothesis testing (Seabold & Perktold, 2010), and scikit-learn provides predictive modeling tools (Pedregosa et al., 2011). Researchers can construct more flexible and study-specific analyses with these packages. Spreadsheet and graphical statistical workflows are also practical alternatives when preprocessing and metadata handling are carefully documented.

NeuroPipe-SABV does not propose a new statistical estimator. Its reason for existing as a separate application is the combination of a guided interface with conventions encountered in behavioral tracking exports. Contributing a header alias or a statistical routine to a general-purpose library would not by itself provide this end-user workflow. The implementation therefore reuses established scientific Python libraries while adding domain-oriented data preparation and presentation. Header aliases inspired by EthoVision XT and ANY-maze exports are included, but compatibility with a vendor's complete range of export formats is not assumed.

The current Panel interface concentrates on data preparation and male–female comparisons. The earlier Streamlit pages additionally contain regression and broader visualization tools. The two interfaces should not be treated as having identical functionality. Studies requiring repeated-measures models, sex-by-treatment interactions, or a planned multiplicity correction need an analysis tailored to that design; separate two-group tests do not answer those questions.

Software design

The architecture separates reusable processing modules in `src/` from interface code. Panel maintains each interactive workflow in an application object, while the older Streamlit implementation uses page-level session state. Both reuse cleaning and feature-transformation helpers. This separation permits processing functions to be tested without a running web server and allows interface migration without replacing the entire data-processing layer.

Header mapping. Column normalization uses exact, fuzzy, and substring matching against a canonical vocabulary. The fuzzy cutoff is 0.85. These permissive matching stages accommodate inconsistent spellings, but can also assign an incorrect meaning to an unfamiliar header. The mapping report records renames, unmatched columns, and collisions; suffixes distinguish colliding names. Researchers must inspect those mappings, especially where different zones, units, or trial identifiers could otherwise be conflated.

Metadata and cleaning. The interface offers a numeric threshold, explicit male/female ID lists, or a female-ID list with other numeric IDs assigned male. These options encode study conventions rather than infer biological sex. The shared routine trims strings while retaining missing measurements and duplicate rows by default. Median imputation requires an explicit selection of measurement columns; identifiers are never imputed. Duplicate removal is separately opt-in. An audit records the final column mapping, individual replacements, removed row positions, and sex assignments. CSV preserves unresolved values, and Excel and JSON exports carry processing provenance. Methods text is generated from the recorded choices. These controls expose preprocessing decisions without establishing that imputation or independent-group inference is appropriate for a particular study.

Quality checks. A reusable artifact-detection module flags velocity extremes, repeated coordinates, coordinate jumps, missing tracking values, and coordinate outliers or positions outside supplied arena bounds. Thresholds are configurable and depend on compatible units and ordered observations. Stationary animals can legitimately have repeated coordinates, so flags identify candidates for inspection rather than confirmed errors. These checks are available through the processing API; they are not automatically enabled in the default Panel processing callback.

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Analysis and export. The Panel interface uses Welch's independent two-sample test and reports descriptive summaries and standardized mean differences. Welch's test relaxes the equal-variance assumption but still requires a defensible independent experimental unit. Matplotlib produces visual summaries (Hunter, 2007). Processed tables can be exported as CSV or Excel, with sex-group summaries in the workbook. Local execution and the supplied container offer alternatives to reliance on a demonstration host, although hosting configuration alone does not establish service availability or production readiness.

Research impact statement

NeuroPipe-SABV was developed in the Laboratory for Comparative Neuropsychology at Towson University in the context of open-field, elevated plus-maze, and Morris water-maze data processing. Its current contribution is a publicly inspectable research beta and a concrete workflow for testing how sex metadata and preprocessing choices affect behavioral summaries.

A committed synthetic open-field example provides eight input records, explicit subject metadata, expected results, and a reproduction script. It includes a missing outcome and an unidentified record. The observed distance comparison has three animals per sex, means of 12 and 18 cm, and Welch t = -3.6742 with four degrees of freedom. A separate reference script computes sample variances and the Welch formula without importing project processing or test helpers; SciPy's t distribution evaluates the tail probability. The reproduction script checks the real Panel callback, statistics, preserved missing values, and CSV/Excel/audit exports. Tests also exercise the legacy Streamlit pages. Deployment instructions, contribution guidance, and a beta-review checklist are included. These materials give prospective users a concrete correctness check before applying the software to their own studies. They support reproducible technical assessment, rather than demonstrating broad external adoption or validation across laboratories. No measured speedup, elimination of transcription errors, or independent certification of scientific validity is claimed here. Further evaluation should compare outputs with reference calculations and representative study datasets, including repeated measurements and exceptions to ID conventions.

AI usage disclosure

Generative AI assisted software implementation and debugging, documentation, and manuscript drafting and revision. In this revision, descriptions were checked against repository source, bibliographic metadata was checked against publisher or bibliographic records, the manuscript was checked for required sections and citation consistency, and synthetic-data tests were executed for the processing modules, real Panel callbacks, and Streamlit pages, including the reported identity and missing-data failure cases. These checks do not replace scientific review or validation of the software. The authors remain responsible for the manuscript and software.

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